Tuan Van Vo

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Education

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI) Abu Dhabi, UAE
PhD in Computer Vision, supervised by Prof. Ian Reid *Sep. 2025 – Sep. 2029 (Expected)*

Vietnam National University Ho Chi Minh City University of Information Technology
Bachelor of Science in Computer Science *Aug. 2017 – June 2021*

- First Class with Honors **(Talented Class – Honors Program)**
- GPA: 8.32/10. **Honors programs** offer special curricula, privileges, scholarships, and recognition for exceptional undergraduate students.

Publications and Under Submissions [Google Scholar]

Publications

[7] Duy Le Dinh Anh, Patrick Amadeus Irawan, **Tuan Van Vo**. Counting to Four is still a Chore for VLMs. In *CVPR 2026 Workshops – The 5th Workshop on Multimodal Foundation Models*, 2026. Paper

[6] Tudor Jianu, Baoru Huang, **Tuan Van Vo**, ..., Olatunji Omisore, Pierre Berthet-Rayne, Sebastiano Fichera, Anh Nguyen. Autonomous Catheterization with Open-source Simulator and Expert Trajectory. In *Handbook of Robotic and Image-Guided Surgery*, 2026. Paper Code

[5] Sy-Tuyen Ho*, **Tuan Van Vo***, Somayeh Ebrahimkhani*, Ngai-Man Cheung. Vision Transformer Neural Architecture Search for Out-of-Distribution Generalization: Benchmark and Insights. In *Conference on Neural Information Processing Systems (NeurIPS)*, 2024. Paper Code. ***First authors.**

[4] **Tuan Van Vo**, Minh Nhat Vu, Baoru Huang, Toan Nguyen, Ngan Le, Thieu Vo, Anh Nguyen. Open-Vocabulary Affordance Detection using Knowledge Distillation and Text-Point Correlation. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2024. **Oral Presentation.** Paper Code Video

[3] Toan Nguyen, Minh Nhat Vu, Baoru Huang, **Tuan Van Vo**, Thuy Tuong Vy Truong, Ngan Le, Thieu Vo, Hoai Bac Le, Anh Nguyen. Language-Conditioned Affordance-Pose Detection in 3D Point Clouds. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2024. **Oral Presentation.** Paper Code Video

[2] **Tuan Van Vo**, Minh Nhat Vu, Baoru Huang, An Vuong, Ngan Le, Thieu Vo, Anh Nguyen. Language-driven Grasp Detection with Mask-guided Attention. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024. **Oral Presentation.** Paper Code

[1] Nam Phuong Nguyen*, **Tuan Van Vo***, Soan T. M. Duong, Chanh D. Tr. Nguyen, Trung Bui, Steven Q. H. Truong. Dual Consistency Assisted Multi-Confident Learning for Hepatic Vessel Segmentation using Noisy Labels. In *British Machine Vision Conference (BMVC)*, 2022. Paper Code. ***First authors.**

Preprints

[10] An Dinh Vuong*, **Tuan Van Vo***, Abdullah Sohail, Haoran Ding, Liang Ma, Xiaodan Liang, Anqing Duan, Ivan Laptev, Ian Reid. World2Act: Latent Action Post-Training via Skill-Compositional World Models. *Under review*, 2026. arXiv Project Page. ***First authors.**

[9] **Tuan Van Vo**, Quang Tan Nguyen, Khang Nguyen, Tran Xuan Nhat, Duy Minh Ho Nguyen, An Thai Le, Vien Anh Ngo, Minh Nhat Vu. ReFineVLA: Multimodal Reasoning-Aware Generalist Robotic Policies via Teacher-Guided Fine-Tuning. *Under review*, 2025. arXiv

[8] Baoru Huang*, **Tuan Van Vo***, Chayun Kongtongvattana, Anh Nguyen. CathAction: A Benchmark for Endovascular Intervention Understanding. *Under review*, 2024. arXiv. ***First authors.**

Patents

[P1] Somayya Ebrahimkhani, Ngai-Man Cheung, BingSheng Jonathan Zheng, Chun Yong Andrew Ngo, Le Hoang Anh, **Tuan Van Vo**, U-Xuan Tan. Machine Learning Model for Plastic Type Classification. *PCT Application WO2026071978A1 (pending)*, 2026. Patent

Academic Services

Reviewer Services - Journal and Conference Reviewer

- Reviewer for TMM-24, ICRA-25, IROS-25, CoRL-25, CoRL-26

Work Experience

Research Lab Experience

Singapore University of Technology and Design, Singapore Sept. 2023 – Mar. 2025

Research Assistant

- Research focus: Zero-Shot Neural Architecture Search
- Vision Transformer Neural Architecture Search for Out-of-Distribution Generalization, **resulting in a publication at NeurIPS 2024**

FPT Software AI Center July. 2022 – June. 2024

AI Research Residency

- Research focus: Multimodal Robot Learning
- **Resulted in two publications at ICRA 2024 and one at IROS 2024**

MMLab - UIT Oct. 2019 – May 2021

Research Assistant, *University of Information Technology (UIT) - VNUHCM*

- Research Thesis: Violence Detection in Video Surveillance Using a Deep Multiple Instance Learning Ranking Approach (PDF)

Industry Experience

VinRobotics, Vingroup Corporation April. 2025 – Aug. 2025

Principal Research Engineer (Founding Team Member) *Vietnam*

- Research focus: Foundation Models for Vision-Language-Action (VLA) in robotics.
- Lead scientist developing large-scale multimodal VLA models that integrate vision, natural language, and action planning for next-generation robotics at VinRobotics.

MoMo (M-Service JSC) June. 2022 – August. 2023

Senior Computer Vision Research Engineer

- Researched and developed AI solutions, specifically designing computer vision models for the eKYC system of the MoMo app
- Designed and implemented Face Anti-Spoofing, Object Detection, OCR, Deep-Fake Detection, Fraud Detection, and Document Understanding models
- Achieved a 99% accuracy rate with low latency, serving nearly 40 million users at about 45 RPS
- **These AI eKYC systems have gained trust and are utilized by the Apple Online Store in Vietnam**

VinBrain (Acquired by NVIDIA) May. 2021 – May. 2022

Applied Scientist

- Conducted research and development of AI for medical segmentation and diagnostic tasks, as well as AI for Smart City and Smart Home solutions
- Designed models for 3D MRI CT-Rectal Tumor Segmentation, Liver Vessel Segmentation, Pet Detection, and Coverface Detection on surveillance cameras
- Successfully delivered all AI models to products to assist doctors. In the surveillance camera project, the AI models were also successfully delivered and deployed at VinHome residential areas for camera-AI based surveillance
- Performed in-depth research and targeted academic papers for prestigious AI conferences, **resulting in a publication at BMVC 2022**

FPT Software May. 2020 – May. 2021

AI Engineer - Technical Lead of Healthcare Projects

- Designed and implemented cutting-edge AI solutions for healthcare in the Japanese market, addressing customer requirements related to model design, logic, evaluation, and reporting
- Designed models for semantic segmentation of dental X-rays and classification of 72 implant types
- Successfully delivered all AI models to the customer

Scholarship and Activities

Full Scholarship and Stipend for PhD Study at Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI)
Rosen Scholarship 2020 & 2021
UIT-Study Scholarships 2018 & 2019
Participated in the ICPC 2019 Vietnam National Programming Contest
Attended and presented two oral papers at ICRA 2024, Yokohama, Japan

Skills

Programming Languages: Python, C/C++, SQL
AI Frameworks and Model Training: PyTorch, TensorFlow, Keras, OpenCV, CUDA basics; experienced in large-scale distributed training of VLA, VLM, LLM, MLLM, Video Generation, and Video World models
Robotics and Deployment: ROS, robotic system deployment, teleoperation-based data collection
Tools and Platforms: Linux, Docker, Google Cloud, Jupyter Notebook/Lab, Git, BigQuery, Oracle
Languages: English, Vietnamese (native)
Typography: LaTeX

References

- **Prof. Ian Reid**
  Department Chair and Professor of Computer Vision at MBZUAI
  Acting Director of Research Administration at MBZUAI
  Email Ian.Reid@mbzuai.ac.ae
  Google Scholar https://scholar.google.com/citations?user=ATkNLcQAAAAJ&hl=en
  Homepage https://mbzuai.ac.ae/study/faculty/ian-reid/

- **Prof. Anh Nguyen**
  Associate Professor at the University of Liverpool
  Director of the Smart Robotic Lab, Department of Computer Science
  Email anh.nguyen@liverpool.ac.uk
  Google Scholar https://scholar.google.com/citations?user=gEbaF0sAAAAJ&hl=en
  Homepage https://www.csc.liv.ac.uk/~anguyen/

- **Prof. Ngai-Man (Man) Cheung**
  Associate Professor at the Singapore University of Technology and Design (SUTD)
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- **Prof. Thieu N. Vo**
  Assistant Professor at the University of Bath
  Department of Computer Science.
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- **Prof. Baoru Huang**
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  Google Scholar https://scholar.google.com/citations?user=unbPvWAAAAAJ&hl=zh-EN
  Homepage https://baoru.netlify.app/